SIMATS ENGINEERING
Department of Information Technology
ITA05 – Computer Vision
ASSIGNMENT 1 – CO1

Assignment Title: Apply, Compare and Justify Image Acquisition, Formation and Processing Techniques for a Textile Fabric Defect Inspection System

CO1: Apply the fundamental concepts, image formation models, and processing techniques for analyzing Computer Vision systems.

Student Name		**Register Number**	
Department / Section	Information Technology	**Academic Year / Semester**	2025–2026 / _____
Faculty Name		**Nature of Assignment**	Individual
Duration	1 Week	**Total Marks**	50
Date of Submission	30.10.2026	**Date of Evaluation**	07.11.2026

Industrial Scenario

Sri Lakshmi Textile Industries, Tiruppur, plans to develop a computer vision-based inspection system for its fabric production line. The system must inspect moving cotton and cotton-polyester fabric, measure fabric width, and identify holes, broken threads, oil stains, weaving irregularities and colour variations. It must estimate the location of each defect and trigger a marking mechanism so that the defective portion can be identified for rework or rejection.

Given Industrial Data and Constraints

Parameter	Specification
Fabric width	1,000 mm
Camera field of view	1,200 × 400 mm
Conveyor speed	0.6 m/s
Smallest defect to be detected	1.5 mm
Oil-stain contrast	5–8 grey levels

Required fabric-width measurement accuracy	±2 mm
Maximum decision time	0.5 s
Distance between camera and marking mechanism	0.3 m
Lighting variation	500–3,000 lux
Camera mounting height	700 mm above the fabric
Available computing system	4-core CPU, 16 GB RAM, no GPU
Software environment	Python, OpenCV and NumPy
Network interface	Gigabit Ethernet
Storage capacity	2 TB
Maximum project budget	₹1,50,000
Image retention period	7 days
Maximum acceptable missed-defect rate	2%

General Instructions

1. Answer all ten tasks.
2. For every task, three candidate techniques or approaches are provided. Analyse all three alternatives, compare their suitability for the given industrial scenario, select the most appropriate option and justify your selection.
3. Support your answers with calculations, diagrams, tables, observations, screenshots or experimental results wherever applicable.
4. Clearly state assumptions and show the steps used in numerical calculations.
5. Your justification must consider the given operating conditions, accuracy, processing time, hardware limitations and budget.
6. Do not provide only definitions. Relate every answer to the textile fabric inspection application.
7. Where implementation is required, use Python, OpenCV and NumPy. Include relevant code snippets and sample output.

TASK 1: Overview and Features of Computer Vision – Compare Grading/Inspection Approaches

Marks: 4

Compare the following three approaches for inspecting fabric defects:

Candidate techniques / approaches:

- Manual inspection by trained workers
- Image-processing-assisted inspection, in which an operator reviews computer-generated results
- Fully automated computer vision inspection

Student tasks:

8. Explain the working principle of each approach.
9. Compare their accuracy, speed, consistency, labour requirement and suitability for continuous fabric production.
10. Select the most suitable approach for Sri Lakshmi Textile Industries and justify your decision using the given production conditions.
11. Identify at least two limitations of your selected approach and suggest how they can be reduced.

TASK 2: Levels of Computer Vision – Pipeline Design and Operation

Marks: 5

Compare the following system designs:

Candidate techniques / approaches:

- Low-level pipeline: image acquisition, preprocessing and enhancement
- Three-level pipeline: low-level processing, intermediate-level feature extraction and high-level defect decision
- Deep-learning-based pipeline: image acquisition, neural-network inference and defect classification

Student tasks:

12. Draw a block diagram for each pipeline.
13. Explain the input and output of each processing stage.
14. Identify the stages required to detect holes, broken threads, stains and weaving irregularities.
15. Compare computational complexity, explainability, hardware requirements and response time.
16. Recommend a pipeline that can operate within the available 4-core CPU system and the 0.5-second decision constraint.

TASK 3: Image Representation Analysis

Marks: 4

Compare the following image representations for fabric inspection:

Candidate techniques / approaches:

- RGB colour representation
- HSV colour representation
- Grayscale representation

Student tasks:

17. Explain how each representation stores image information.
18. Discuss the suitability of each representation for identifying colour variations, oil stains, holes and broken threads.
19. Compare memory requirements and processing complexity.

20. Choose a suitable representation or combination of representations and justify your choice with examples from the scenario.

TASK 4: Connectivity Analysis

Marks: 4

Compare the following pixel-connectivity approaches for identifying defect regions:

Candidate techniques / approaches:

- 4-connectivity
- 8-connectivity
- m-connectivity (mixed connectivity)

Student tasks:

21. Define each connectivity approach with a suitable pixel-neighbour diagram.
22. For a small binary image containing diagonally and horizontally connected defect pixels, identify the connected components using all three methods.
23. Explain how the choice of connectivity may affect defect-region size and component counting.
24. Recommend the most appropriate connectivity method for detecting small holes and irregular fabric defects.

TASK 5: Image Acquisition Analysis

Marks: 5

Compare the following camera acquisition options for the moving fabric line:

Candidate techniques / approaches:

- Rolling-shutter camera
- Global-shutter camera
- Line-scan camera

Student tasks:

25. Explain the image acquisition principle of each camera type.
26. Analyse the effect of fabric movement at 0.6 m/s on image distortion and defect visibility.
27. Compare image quality, suitability for continuous web inspection, installation complexity and cost.
28. Select a camera type for the given fabric width and production process.
29. Justify the selection with reference to the 1.5 mm minimum defect size and 0.5-second decision requirement.

TASK 6: Sampling Analysis

Marks: 5

Evaluate the following image sampling resolutions:

Candidate techniques / approaches:

- Low resolution: 1280 × 1024 pixels
- Medium resolution: 2448 × 2048 pixels
- High resolution: 4096 × 3000 pixels

Student tasks:

30. Calculate the total number of pixels for each resolution.
31. Estimate the approximate spatial sampling interval across the 1,200 mm field of view, assuming the image width corresponds to the full field-of-view width.
32. Determine whether a 1.5 mm defect can be represented adequately at each resolution.
33. Compare image storage, processing time and defect-detection capability.
34. Select a resolution that balances defect visibility and the available computing resources. Show all calculations.

TASK 7: Quantization Analysis

Marks: 4

Compare the following intensity quantization options:

Candidate techniques / approaches:

- 8-bit quantization
- 12-bit quantization
- Reduced 4–6 bit quantization

Student tasks:

35. Calculate the number of intensity levels available in each option.
36. Explain how quantization affects the representation of low-contrast oil stains having only 5–8 grey-level differences.
37. Compare image storage requirements and processing implications.
38. Recommend a quantization level for reliable defect detection under the stated lighting conditions. Justify your answer.

TASK 8: Image Formation Analysis

Marks: 5

Compare the following image formation and calibration approaches:

Candidate techniques / approaches:

- Standard perspective imaging without calibration
- Calibrated imaging with geometric correction
- Controlled-illumination imaging with suitable calibration

Student tasks:

39. Explain the image formation process for a camera mounted 700 mm above the fabric.
40. Discuss perspective distortion and its effect on fabric-width measurement.
41. Explain how camera calibration can improve measurement accuracy.
42. Analyse how lighting variation from 500 to 3,000 lux can affect defect visibility.
43. Recommend an image formation and illumination arrangement that can achieve the required width accuracy of ±2 mm.

TASK 9: Implementation and Performance Evaluation

Marks: 6

Design and compare three complete implementation configurations:

Candidate techniques / approaches:

- Budget configuration: minimum-cost camera, basic illumination and CPU-based processing
- Balanced configuration: suitable camera, controlled illumination and optimised CPU-based processing
- Premium configuration: high-resolution industrial camera, advanced illumination and higher-performance processing hardware

Student tasks:

44. Prepare a block diagram showing camera, illumination, processing unit, storage, network and marking mechanism.
45. Prepare a comparison table covering estimated cost, resolution, processing time, storage and expected inspection capability.
46. Explain how the system will detect and record holes, broken threads, stains and weaving irregularities.
47. Describe how the system will communicate the defect location to the marking mechanism located 0.3 m downstream.
48. Evaluate whether each configuration can meet the 0.5-second decision time, 7-day image retention and 2% maximum missed-defect rate.
49. Select one configuration that fits the ₹1,50,000 budget and justify your selection. Clearly state any assumptions used in cost estimates.

TASK 10: Front-End Adaptation Challenge

Marks: 8

Assume that the textile company receives a new export order with the following changed requirements: the system must detect defects as small as 0.4 mm, identify stains visible under ultraviolet (UV) illumination, operate at a conveyor speed of 1.2 m/s, inspect twice the current production volume and produce a decision within 0.3 seconds. The existing PC and project budget must remain unchanged.

Candidate techniques / approaches:

- Retain the existing system with only software changes

- Modify the existing system by upgrading selected components
- Redesign the inspection system using a different acquisition and processing architecture

Student tasks:

50. Identify which existing system components will become inadequate under the new requirements.
51. Analyse the effect of the increased conveyor speed and smaller defect size on image sampling and acquisition.
52. Explain how UV illumination or an additional imaging channel could help identify the new stain type.
53. Compare the three adaptation strategies in terms of feasibility, accuracy, processing time, cost and implementation risk.
54. Recommend the most practical strategy under the unchanged PC and budget constraints.
55. Prepare an updated system block diagram and list the changes required.
56. Clearly identify any requirements that cannot realistically be achieved without additional resources, and justify your conclusion.

Mark Distribution

Task	Topic	Marks
1	Overview and Features of Computer Vision	4
2	Levels of Computer Vision	5
3	Image Representation	4
4	Connectivity Analysis	4
5	Image Acquisition	5
6	Sampling Analysis	5
7	Quantization Analysis	4
8	Image Formation	5
9	Implementation and Performance Evaluation	6
10	Front-End Adaptation Challenge	8
	TOTAL	50

Submission Requirements

- Submit a neatly formatted assignment report with task-wise headings.
- Include hand-drawn or computer-generated diagrams wherever required.
- Show all numerical calculations with formulae, substitution and final units.
- Include comparison tables and evidence supporting the selected techniques.

- For implementation tasks, attach source code, sample input images and output screenshots.
- Mention references, tools and software libraries used.